THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · EASY LEVEL UNIT TEST SET 2

Chapters 8–11: **Introduction to Trigonometry, Some Applications of Trigonometry, Circles, Areas Related to Circles** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The area of a circle is 154 cm^2. The radius of the circle is: 1

(a) 24.5 cm **(b)** 49 cm **(c)** 14 cm **(d)** 7 cm

2. $\sin 25^\circ = \cos$ of: 1

(a) 65° **(b)** 155° **(c)** 40° **(d)** 25°

3. From the top of a cliff 12 m high, the angle of depression of a boat on the sea is 45°. The horizontal distance of the boat from the foot of the cliff is: 1

(a) 6 m **(b)** $12\sqrt{3}$ m **(c)** 12 m **(d)** 24 m

4. PA and PB are tangents from an external point P to a circle with centre O, touching it at A and B. $\angle AOB = 110^\circ$. The $\angle APB$ is: 1

(a) 220° **(b)** 35° **(c)** 70° **(d)** 110°

5. A sector of a circle of radius 21 cm has central angle 60°. The area of the sector is: 1

(a) 462 cm² **(b)** 1386 cm² **(c)** 231 cm² **(d)** 22 cm²

6. $\sec\theta$ is the reciprocal of: 1

(a) $\cot\theta$ **(b)** $\cos\theta$ **(c)** $\sin\theta$ **(d)** $\tan\theta$

7. A ladder 8 m long leans against a vertical wall and makes an angle of 60° with the level ground. The height reached on the wall is: 1

(a) 4 m **(b)** $2\sqrt{3}$ m **(c)** $8\sqrt{3}$ m **(d)** $4\sqrt{3}$ m

8. PA is a tangent at A to a circle with centre O, and $\angle OPA = 30^\circ$. The $\angle AOP$ is: 1

(a) 150° **(b)** 120° **(c)** 30° **(d)** 60°

9. A sector of a circle of radius 21 cm has central angle 60°. The length of its arc is: 1

(a) 231 cm **(b)** 44 cm **(c)** 64 cm **(d)** 22 cm

10. In a right triangle with perpendicular 12, base 5 and hypotenuse 13, $\sin\theta =$ 1

(a) $\frac{5}{13}$ **(b)** $\frac{12}{5}$ **(c)** $\frac{12}{13}$ **(d)** $\frac{13}{12}$

11. The line drawn from the eye of an observer to the point of the object being viewed is called the: 1

(a) line of sight **(b)** horizontal line **(c)** tangent **(d)** normal

12. A tangent to a circle meets the circle in exactly: 1

(a) two points **(b)** three points **(c)** one point **(d)** no point

13. A semicircular protractor-shaped plate has radius 7 cm. The perimeter of the plate is: 1

(a) 29 cm **(b)** 22 cm **(c)** 58 cm **(d)** 36 cm

14. In a right triangle with perpendicular 12, base 5 and hypotenuse 13, $\tan\theta =$ 1

(a) $\frac{12}{13}$ **(b)** $\frac{12}{5}$ **(c)** $\frac{5}{13}$ **(d)** $\frac{5}{12}$

15. From a point on level ground, 12 m away from the foot of a vertical tower, the angle of elevation of its top is 45°. The height of the tower is: 1

(a) 24 m **(b)** $12\sqrt{3}$ m **(c)** $4\sqrt{3}$ m **(d)** 12 m

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. A vertical pole is 14 m high. The angle of elevation of the sun is 45° and the pole casts a shadow on level ground. Find the length of the shadow. 2

17. A tangent to a circle of radius 4 cm has length 3 cm from the external point P to the point of contact. Find the distance of P from the centre of the circle. 2

18. The radius of a circle is 3.5 m. Find the circumference of the circle. 2

19. Evaluate: $\sin^2 25^\circ + \cos^2 25^\circ$. 2

20. The string of a kite is 20 m long. It is taut and makes an angle of 30° with the level ground. Find the height of the kite above the ground. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. A sector of a circle of radius 21 cm has central angle 60°. Find the perimeter of the sector. 3

22. Evaluate: $\sin^2 30^\circ + \cos^2 60^\circ$. 3

23. A vertical pole is $12\sqrt{3}$ m high. The angle of elevation of the sun is 60° and the pole casts a shadow on level ground. Find the length of the shadow. 3

24. PA is a tangent at A to a circle with centre O, and $\angle OPA = 20^\circ$. Find the $\angle AOP$. 3

OR

A semicircle has radius 7 cm. Find the area of the semicircle. 3

25. In $\triangle ABC$, right-angled at B, $AB = 5$ cm and $BC = 12$ cm. Find $\sin A$, $\cos A$ and $\tan A$. 3

OR

A ladder 16 m long leans against a vertical wall and makes an angle of 60° with the level ground. Find the height reached on the wall and the distance of the foot of the ladder from the wall (in this order). 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. In $\triangle PQR$, right-angled at Q, $PQ = 6$ cm and $QR = 8$ cm. Find $\sin P$, $\cos P$, $\tan P$ and check that $\sin^2 P + \cos^2 P = 1$. 5

OR

The angle of elevation of the top of a vertical tower from a point on level ground is 30°. After walking 6 m towards the foot of the tower along a straight line, the angle of elevation becomes 60°. Find the height of the tower and the initial distance of the point from its foot (in this order). 5

27. Two concentric circles have radii 25 cm and 7 cm. A chord of the larger circle touches the smaller circle. Find the length of the chord. 5

OR

The largest possible circle is cut from a square sheet of side 14 cm. Find the area of the sheet left over. 5

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND